Why Edge Computing is Important for IoT Applications

Smart IoT devices are all around us today, and are growing exponentially each year. With this growth come challenges in managing the data generated by these devices. Edge computing comes to the rescue here, as it processes data closer to the IoT device and helps to generate real-time responses.

IoT devices generate a very large amount of data that is processed and analysed for use in various applications. In the early days of IoT, most devices sent this data to the cloud for storage and analysis. As cloud servers are located very far from IoT devices, this led to a delay in getting processed data. New IoT applications require real-time data processing, for which cloud computing is too costly and not fast enough. Edge computing brings data processing services near the devices or the source of the data. This allows data from various IoT devices to be processed and analysed at the edge of the network, which results in lower latency and enhanced efficiency in data transportation. Data can then be sent to the cloud or data centre for long term processing and storage.

The Internet of Things or IoT refers to the large volume of intelligent devices connected to the Internet. These devices can be called nodes; they sense, collect and send data to each other through various communication protocols. IoT devices collect a large volume of heterogeneous data generated by sensors, which needs processing for extracting useful information to offer services to users.

In traditional computing, data gathered by IoT devices is sent to cloud servers for processing, and the results are transferred to IoT devices for necessary action. The time required to send and get processed data from the cloud server is more, which is not acceptable for real-time applications such as healthcare, smart transportation, and smart grids.

The challenges and emerging applications of IoT

As the data generated by various IoT applications increases exponentially, it brings new challenges in managing